

Math Class 9th Federal Board

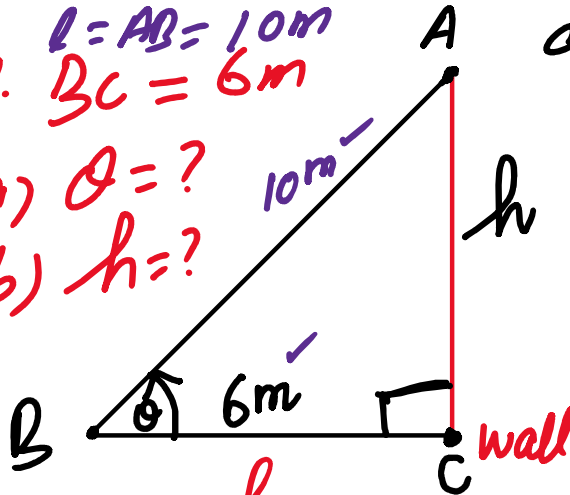
Chapter 06 Trigonometry and

Bearing Part-II

Q2(16). A 10 m long ladder leans against a vertical wall. Its foot is 6 m from the base of the wall. Find: (a) the angle the ladder makes with the ground, (b) the height at which the ladder touches the wall. [4 marks]

Sol:- Given: $AB = 10\text{m}$
 $BC = 6\text{m}$

To find: a) $\theta = ?$
b) $h = ?$



a)

$$\cos \theta = \frac{6\text{m}}{10\text{m}}$$

$$\theta = \cos^{-1}\left(\frac{3}{5}\right)$$

$$\boxed{\theta = 53.13^\circ}$$

b) $\tan \theta = \frac{h}{6}$

$$\tan 53.13^\circ = \frac{h}{6\text{m}}$$

$$\Rightarrow h = 6 \times \tan 53.13^\circ \text{m}$$

$$\boxed{h = 8\text{m}}$$

Thus, $\boxed{\theta = 53.13^\circ, h = 8\text{m}}$

Q2(17). Find the value of a in the following figure. [4 marks]

Sol: In right-angled
 $\triangle ABD$
By Pythagorean
Therm.

$$16^2 = \overline{BD}^2 + 10^2$$

$$\overline{BD}^2 = 16^2 - 10^2$$

$$\overline{BD} = \sqrt{16^2 - 10^2} \Rightarrow \overline{BD} = 2\sqrt{39}$$

$$\Rightarrow \boxed{\overline{BD} \approx 12.49}$$

In right-angled $\triangle ABC$,

$$14^2 = 10^2 + \overline{BC}^2$$

$$\overline{BC} = \sqrt{14^2 - 10^2}$$

$$\boxed{\overline{BC} = 4\sqrt{6} \approx 9.80}$$

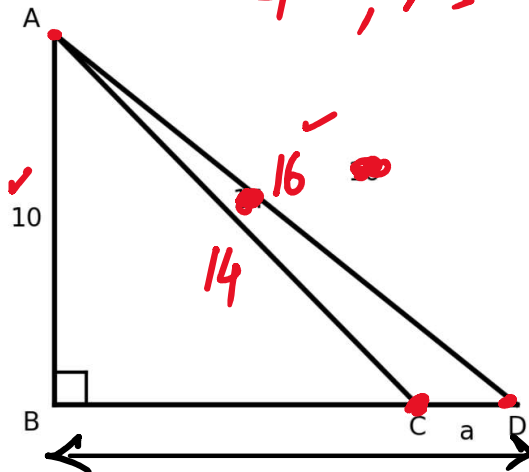
$$\text{As } a = \overline{BD} - \overline{BC}$$

$$a = 2\sqrt{39} - 4\sqrt{6}$$

$$\boxed{a \approx 2.69} \quad \text{Ans}$$

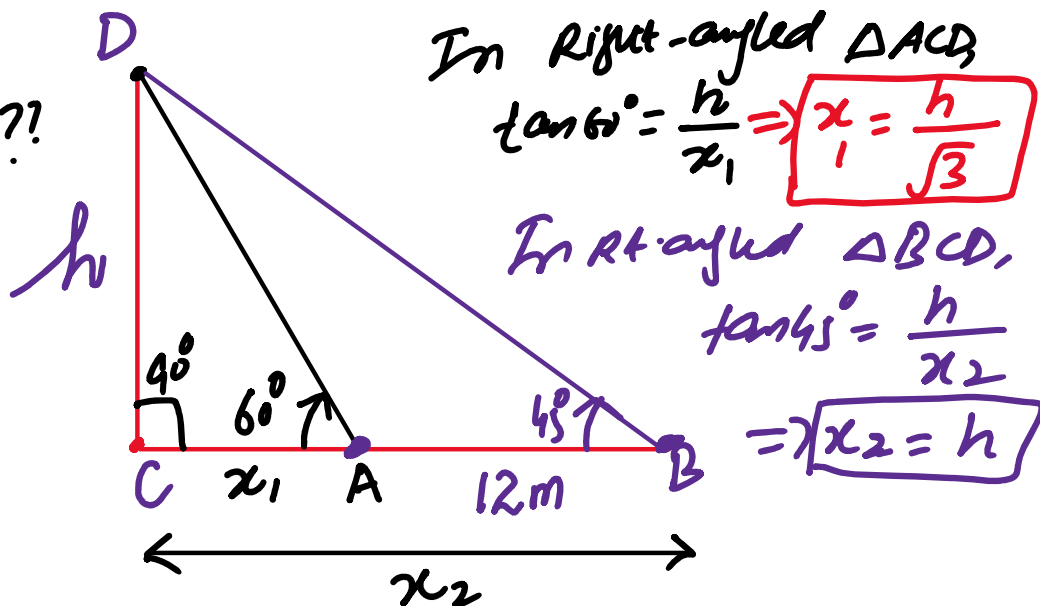
Given, $\overline{AB} = 10$, $\overline{AC} = 14$,
 $\overline{AD} = 16$

To Find:
 $a = ?$



Q2(18). If the shadow of a building increases by 12 m when the angle of elevation of the sun decreases from 60° to 45° , find the height of the building. [4 marks]

Sol: $h = ?$



Given Condition $x_1 + 12 = x_2$

$$\Rightarrow x_2 - x_1 = 12$$

$$h - \frac{h}{\sqrt{3}} = 12$$

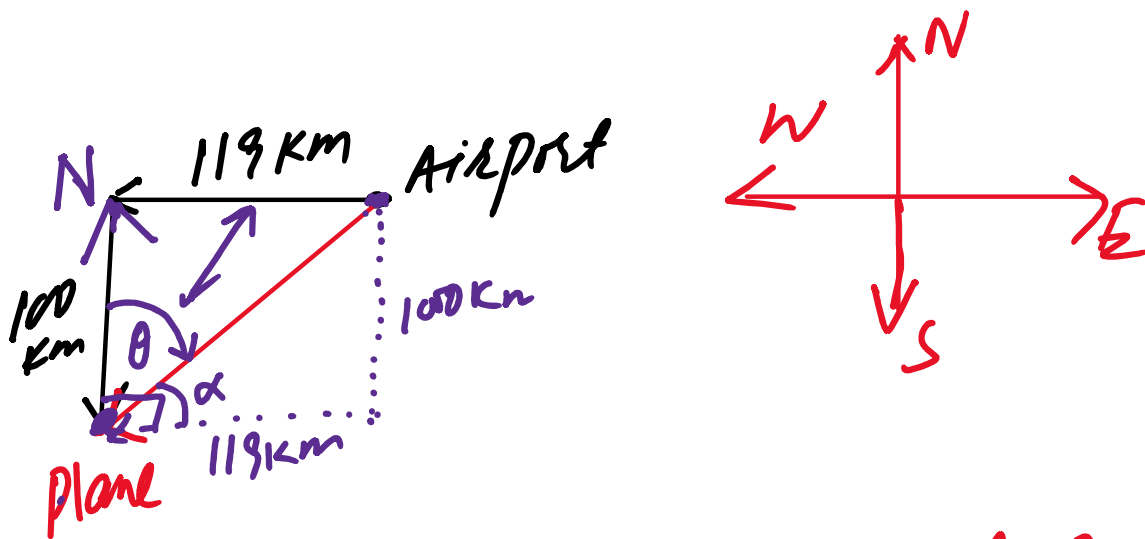
$$h \left(1 - \frac{1}{\sqrt{3}} \right) = 12$$

$$h = \frac{12}{1 - \frac{1}{\sqrt{3}}} = 18 + 6\sqrt{3}$$

$$\Rightarrow \boxed{h = 28.39 \text{ m}}$$

Q2(19). A plane is 119 km west and 100 km south of an airport. At what bearing should the pilot fly directly back to the airport? [4 marks]

... $\uparrow N$



To fly back to the Airport,
plane must cover 119 km
East & 100 km North.

$$\tan \theta = \frac{119 \text{ km}}{100 \text{ km}}$$

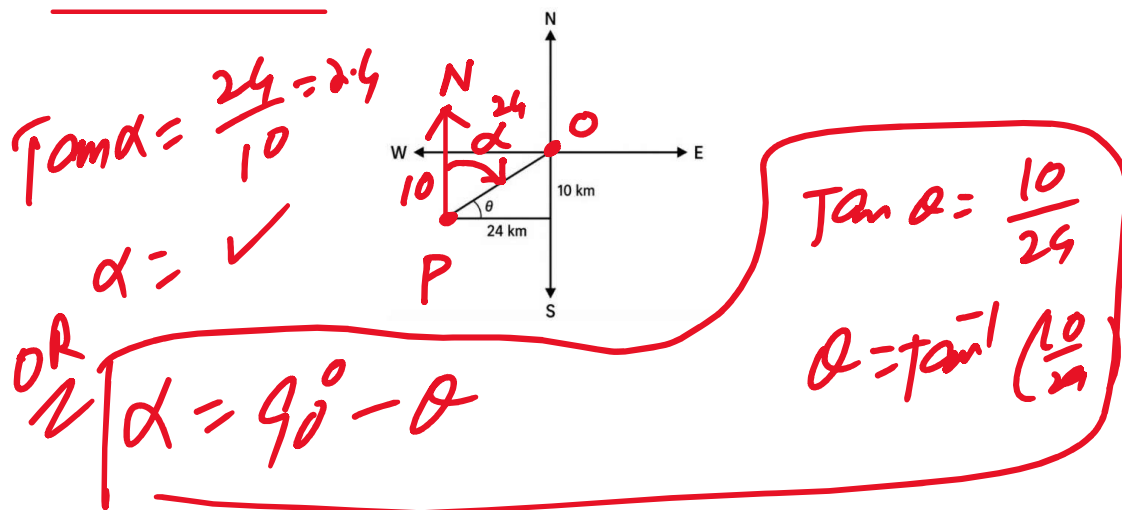
$$\theta = \tan^{-1}\left(\frac{119}{100}\right) \approx 49.96^\circ \approx 50^\circ$$

The plane will have bearing

$$\text{N } 49.96^\circ \text{ E}$$

or 050°

Q2(6). A cyclist travels 10 km due south and then 24 km due west. What simple bearing should the cyclist take to return directly to the starting point? [4 marks]



Q2(20). Prove that $\sqrt{[(1 - \cos \theta)/(1 + \cos \theta)]} = \sin \theta / (1 + \cos \theta)$. [4 marks]

Sol: L.H.S = $\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$

$(1 - \cos \theta)^2$
 $1 - \cos^2 \theta = \sin^2 \theta$

$$= \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta} \times \frac{1 + \cos \theta}{1 + \cos \theta}}$$

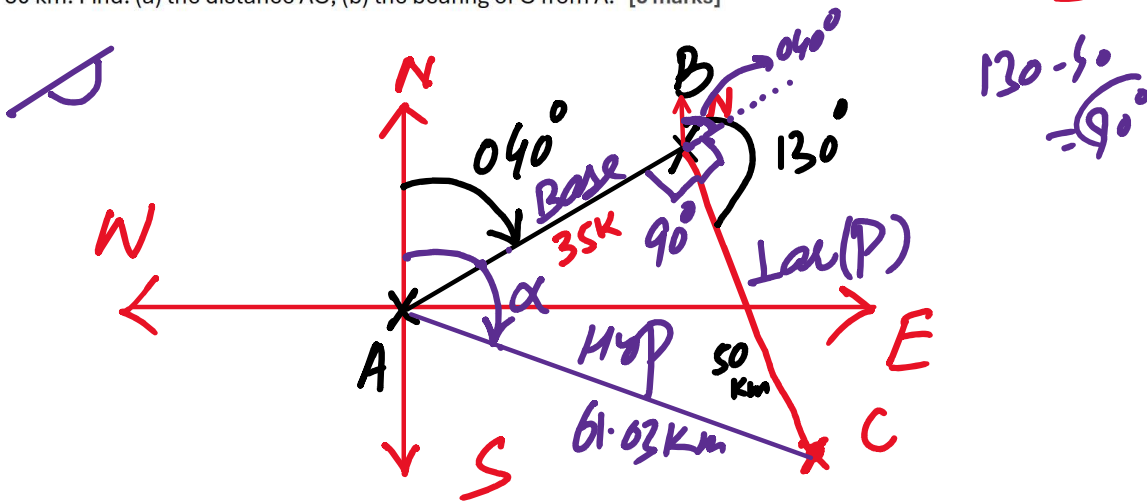
$$= \frac{\sqrt{1 - \cos^2 \theta}}{\sqrt{(1 + \cos \theta)^2}}$$

$$= \frac{\sqrt{\sin^2 \theta}}{1 + \cos \theta}$$

$$= \sin \theta = \text{R.H.S}$$

$\frac{\sin \alpha}{1 + \cos \alpha} = \text{R.H.S}$

Q7. A ship sails from port A to port B on a bearing of 040° for 35 km. It then sails from B to C on a bearing of 130° for 50 km. Find: (a) the distance AC, (b) the bearing of C from A. [8 marks]



a) AC ?

As $\angle ABC = 90^\circ$
In right-angled $\triangle ABC$

$$AC^2 = AB^2 + BC^2$$

$$AC = \sqrt{(35\text{ km})^2 + (50\text{ km})^2}$$

$$\boxed{AC = 61.03 \text{ km}}$$

b) Bearing of C from A

$$= 040^\circ + \alpha$$

$$= 040^\circ + \tan^{-1}\left(\frac{50\text{km}}{35\text{km}}\right)$$

($\because \triangle ABC$ is a right angled $\triangle$)

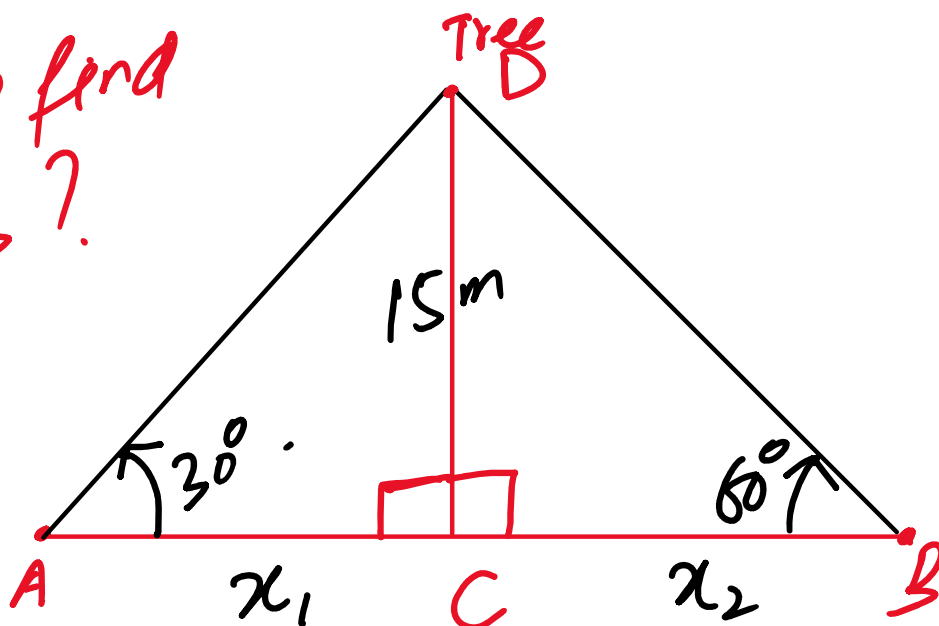
$$= 040^\circ + 055^\circ$$

$$= \boxed{095^\circ} \text{ Ans}$$

Thus Bearing from A to C
 $\equiv 095^\circ$

Q3. Two men on opposite sides of a tree are in line with it. The angles of elevation of the top of the tree are 30° and 60°. The height of the tree is 15 m. Find the distance between the men. [8 marks]

Sol:- To find
AB?



$$\text{Let } AB = x = AC + BC$$

$$AC = x_1, \quad BC = x_2$$

$$\Rightarrow \boxed{x = x_1 + x_2} \rightarrow \textcircled{1}$$

In right-angled $\triangle ACD$,

$$\tan 30^\circ = \frac{15\text{m}}{x_1}$$

$$\Rightarrow x_1 = \frac{15\text{m}}{\tan 30^\circ}$$

$$\boxed{x_1 = 15\sqrt{3}\text{m}}$$

In right-angled $\triangle BCD$,

$$\tan 60^\circ = \frac{15\text{m}}{x_2}$$

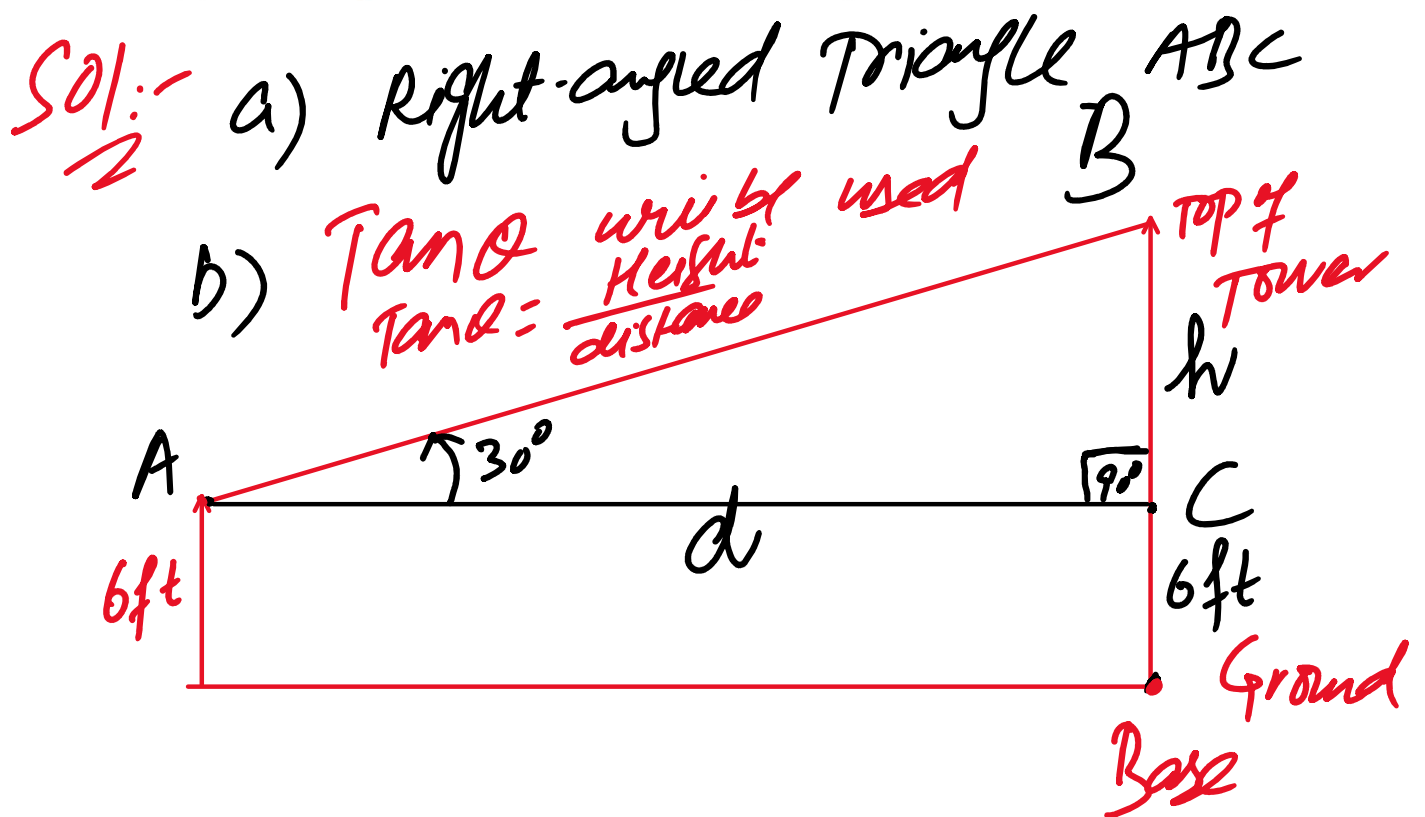
$$\Rightarrow \boxed{x_2 = \frac{15\text{m}}{\sqrt{3}}}$$

Put values of x_1 & x_2 in (1), we get

$$x = 15\sqrt{3} \text{ m} + \frac{15}{\sqrt{3}} \text{ m}$$

$$x = 90\sqrt{3} \text{ m} \approx \cancel{154.7} \text{ m} \\ \approx 34.64 \text{ m}$$

Q4. From the eye level of a 6-ft-tall person standing on the ground, the angle of elevation of the top of a tower is 30° . (a) What type of triangle is formed? (b) Which trigonometric ratio would be used to find the height of the tower? (c) Find the height of the tower from its base. [8 marks]



c) Height of the tower

$$c) \text{ Height of tower} = 6 \text{ ft} + h \rightarrow \textcircled{1}$$

In right-angled $\triangle ABC$,

$$\tan 30^\circ = \frac{h}{d}$$

($\because d = \text{Horizontal distance b/w person \& Tower}$)

$$h = \tan 30^\circ \times d$$

$$h = \frac{d}{\sqrt{3}} \text{ ft}$$

$$\text{So, Eqn. (1)} \Rightarrow \text{Height of the tower} = 6 \text{ ft} + \frac{d}{\sqrt{3}} \text{ ft}$$

Q5. The tide height is modeled by $H = H_0 + A \sin(2\pi t/T)$, where H_0 is average tide height, A is amplitude, t is time in hours, and T is the period. If $H_0 = 2$ m, $A = 1$ m, $T = 24$ h, find H at $t = 0, 6$ and 18 hours. [8 marks]

The height H of the tide at a coastal location varies over a day, modeled by $H = H_0 + A \cdot \sin\left(\frac{2\pi t}{T}\right)$, with H_0 : the average tide height, A : the amplitude of tidal variation, t : the time in hours, and T : the period of tidal cycle in hours. If $H_0 = 2$ m, $A = 1$ m, $T = 24$ hours use trigonometry to find the tide's height at $t = 0, 6, 18$ hours.

8

Given:

$$H = H_0 + A \sin\left(\frac{2\pi t}{T}\right)$$

$$H = 2 + \sin\left(\frac{2\pi t}{24}\right) \text{ m} \rightarrow \textcircled{1}$$

i) $t = 0$ hours

$$H = 2 + \sin\left(\frac{2\pi \times 0}{24}\right) \text{ m}$$

$$H = (2 + 0) \text{ m} \Rightarrow \boxed{H = 2 \text{ m}}$$

ii) for $t = 6$ hours

$$H = 2 + \sin\left(\frac{2\pi \times 6}{24}\right) \text{ m}$$

$$H = 2 + \sin\left(\frac{\pi}{2}\right) \text{ m}$$

$$H = (2 + 1) \text{ m} \rightarrow \boxed{H = 3 \text{ m}}$$

$$\Rightarrow \underline{H = 3m}$$

iii) for $t = 18$ hours.

$$H = 2 + \sin\left(\frac{2\pi \times 18}{24 \times 12}\right) \text{ m}$$

$$H = 2 + \sin\left(\frac{3\pi}{2}\right) \text{ m}$$

$$H = (2 - 1) \text{ m}$$

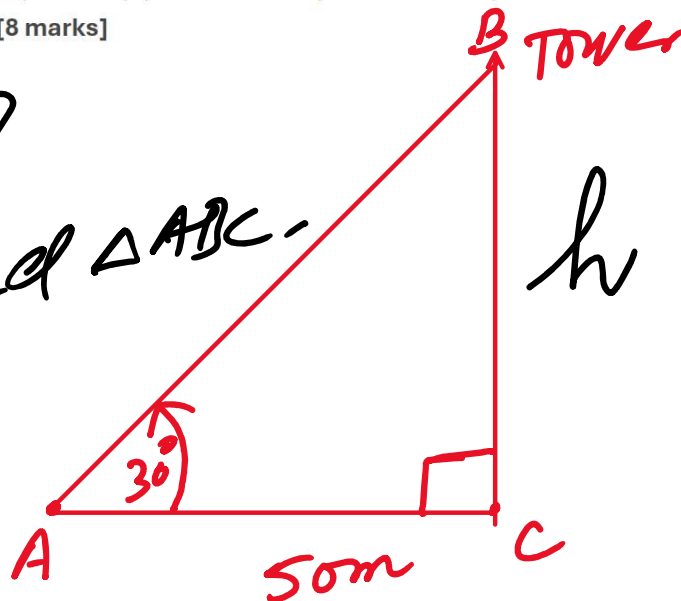
$$\boxed{H = 1 \text{ m}}$$

Q6. A person sees the top of a tower at an angle of elevation of 30° . The horizontal distance to the tower is 50 m.
(a) Find the tower height to two decimal places. (b) A bird directly above the top is seen at 60° . Find the vertical distance of the bird from the ground. [8 marks]

Sol: a) $h = ?$

In right-angled $\triangle ABC$.

$$\tan 30^\circ = \frac{h}{50 \text{ m}}$$

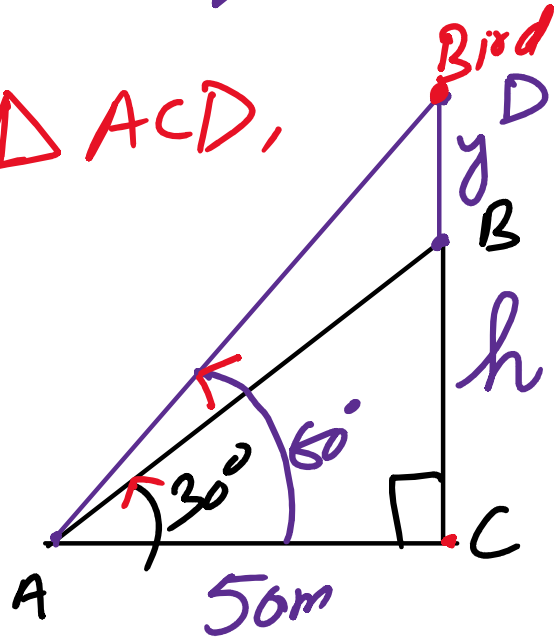


$$\Rightarrow h = \frac{1}{\sqrt{3}} \times 50m$$

$$h = 28.87m$$

b) In rt-angled $\triangle ACD$,

$$\tan 60^\circ = \frac{CD}{50m}$$



$$CD = 50m \times \sqrt{3}$$

$$CD = 86.6m$$

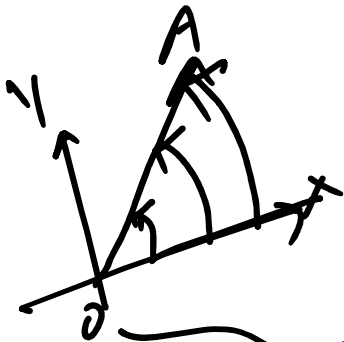
→ Height of the Bird
from the Ground.

c) Find the height of the
Bird from the top of the
... the Bird

Power to the Bird
pos →

$$y = CD - h$$

$$= 86.60\text{m} - 28.57\text{m}$$



Q8. An aeroplane at an altitude of 900 m observes two ships sailing towards it in the same direction. Their angles of depression are 30° and 60° . Find the distance between the ships. [8 marks]

Sol: To find
= $x = BC = \text{distance}$

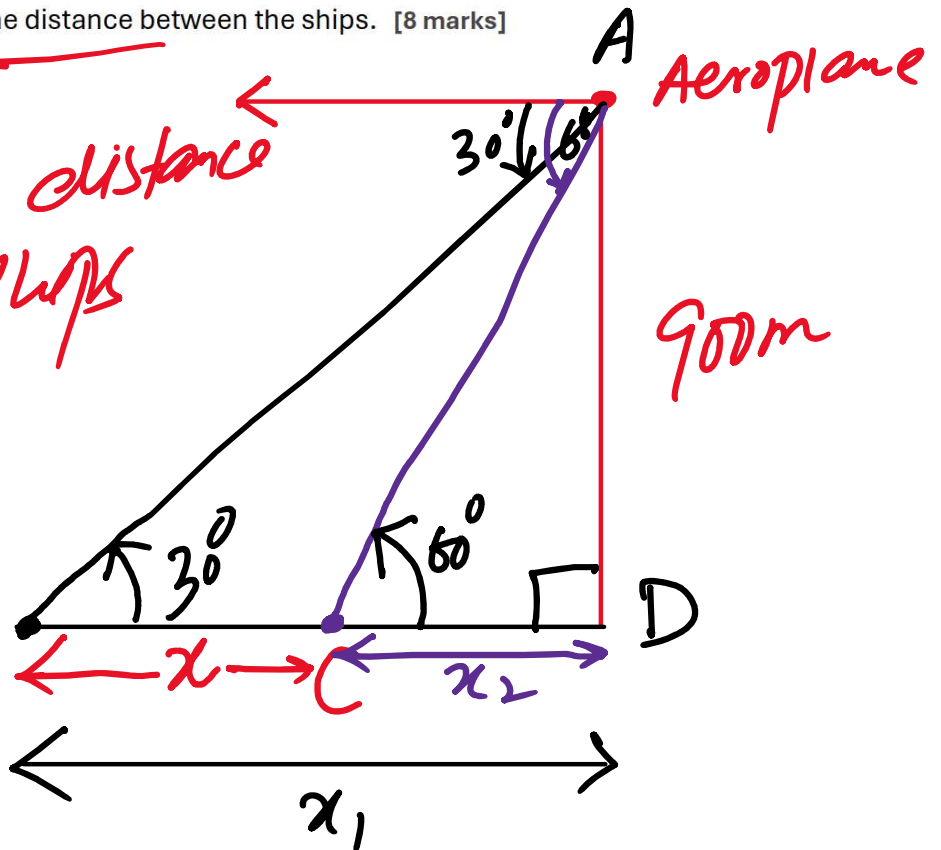
b/w Two ships

et. etc

In $\triangle ABD$,

$$\tan 30^\circ = \frac{900\text{m}}{x_1}$$

$$x_1 = \frac{900\text{m}}{\frac{1}{\sqrt{3}}}$$



$\sqrt{3}$

$$x_1 = 900\sqrt{3} \text{ m} = 1558.85 \text{ m}$$

In rt-angled $\triangle ACP$,

$$\tan 60^\circ = \frac{900}{x_2}$$

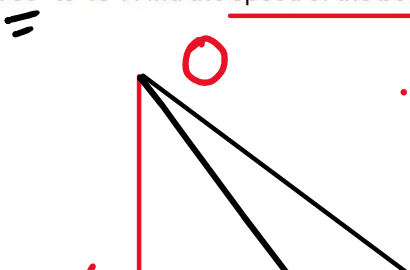
$$x_2 = \frac{900}{\sqrt{3}} = 519.62 \text{ m}$$

As $x = x_1 - x_2$

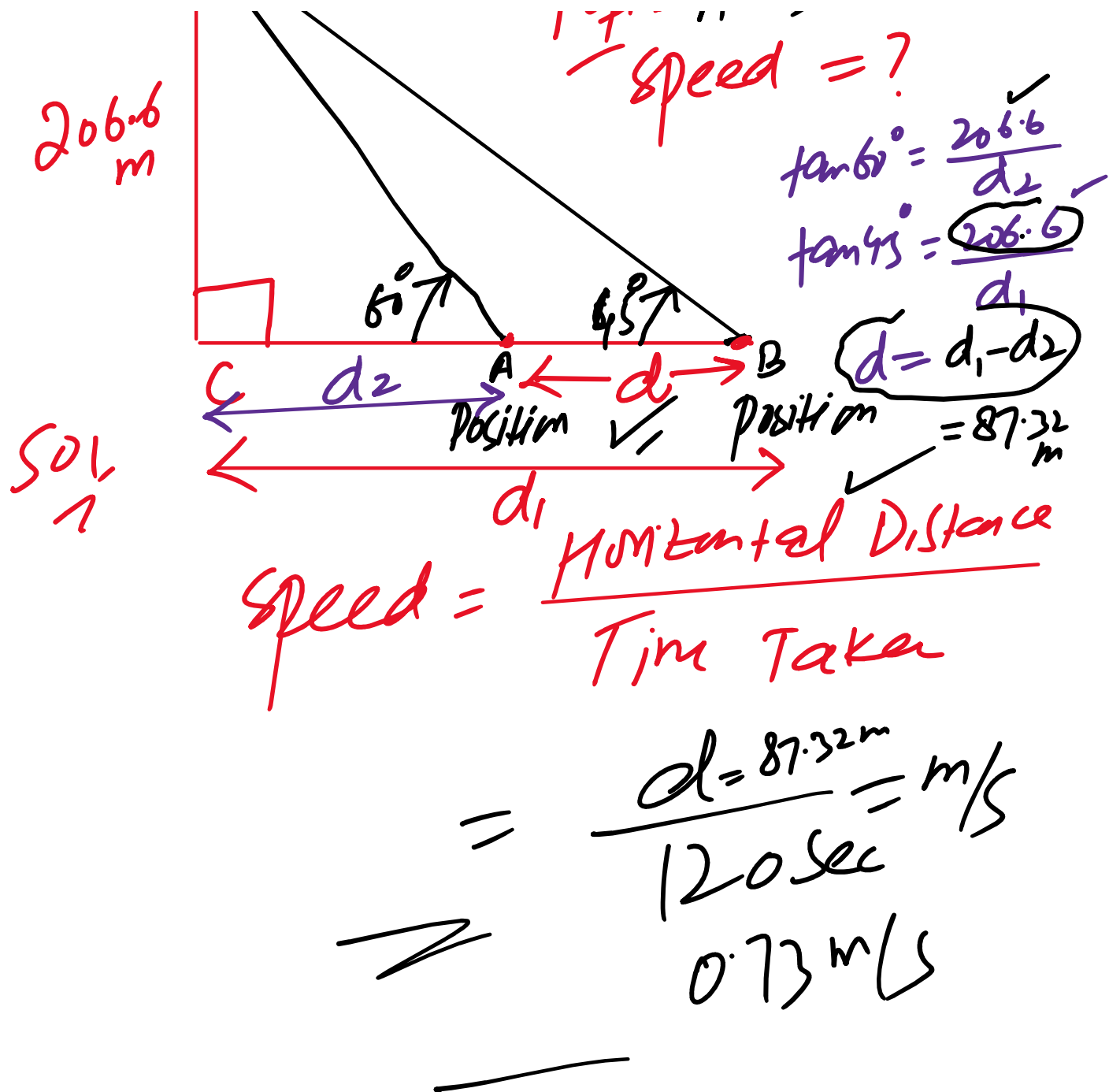
$$x = 1558.85 - 519.62$$

$$x = 1039.23 \text{ m}$$

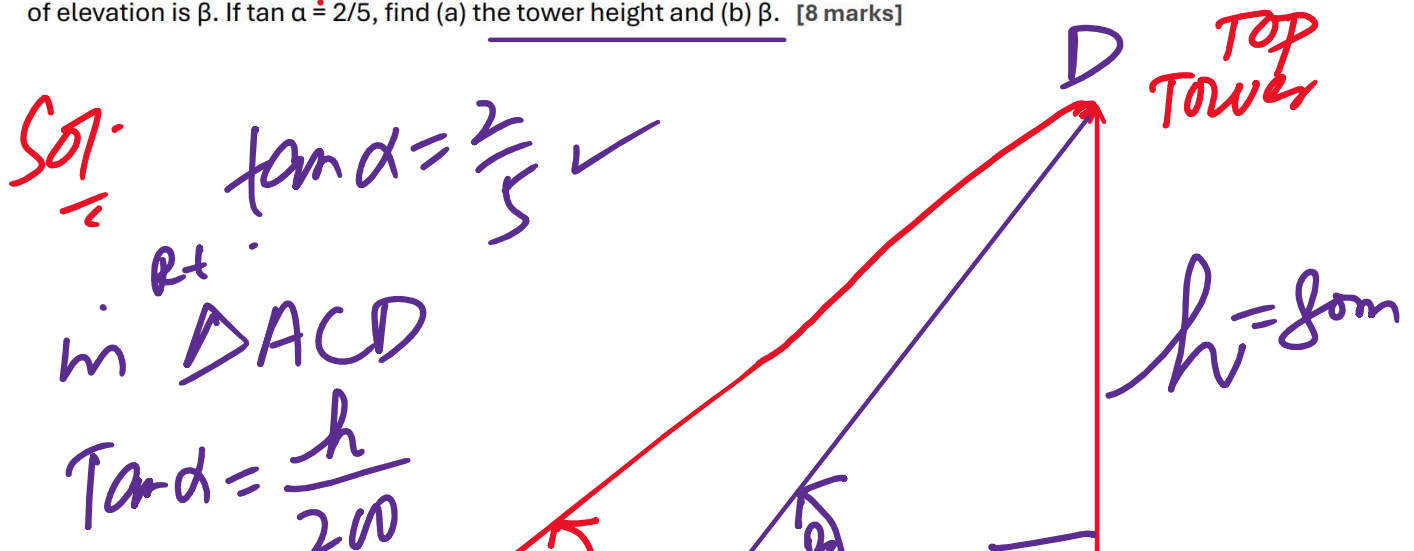
Q9. A boat moving away from a lighthouse 206.6 m high takes 120 seconds for the angle of elevation of the top to decrease from 60° to 45° . Find the speed of the boat. [8 marks]



Given Time taken for
to find $A \rightarrow B = 120 \text{ s}$
Speed = ?

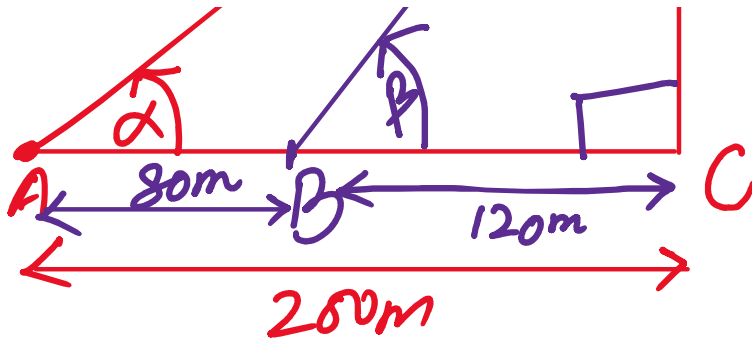


Q10. From point A, 200 m from a tower, the angle of elevation is α . Point B is 80 m nearer the tower, and its angle of elevation is β . If $\tan \alpha = \frac{2}{5}$, find (a) the tower height and (b) β . [8 marks]



$$\tan \alpha = \frac{h}{200}$$

$$\frac{2}{5} = \frac{h}{200}$$

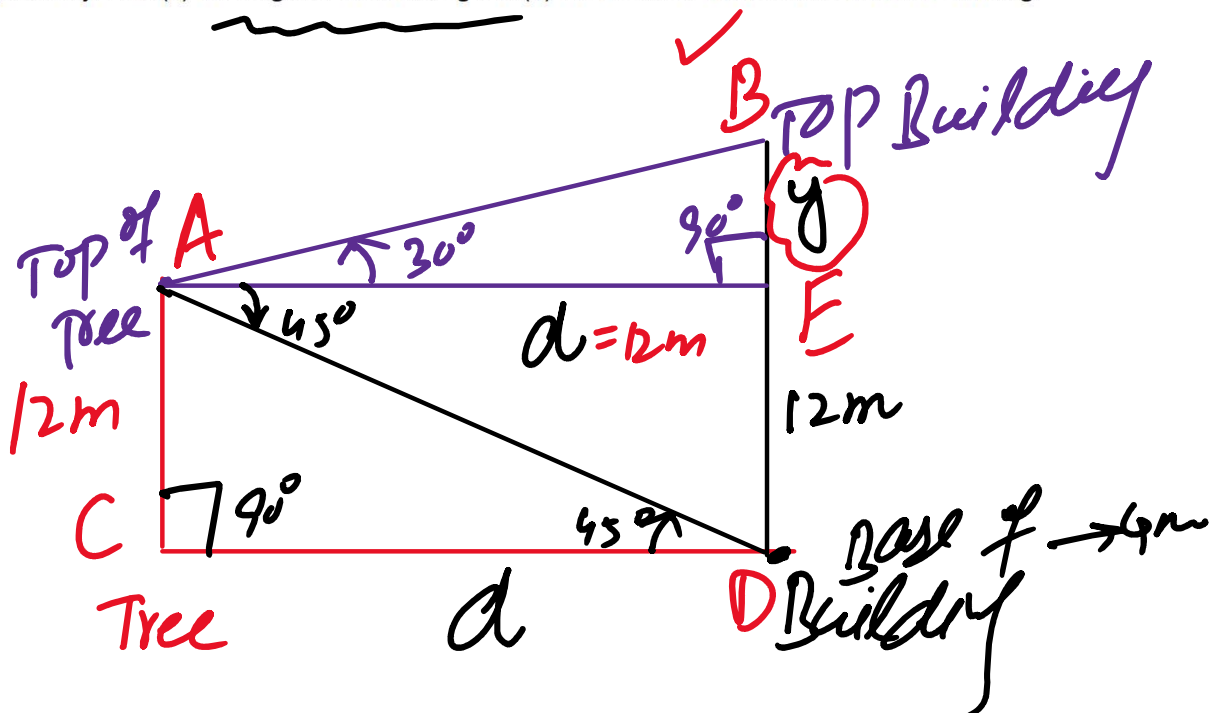


$$h = \frac{400}{5} \Rightarrow \boxed{h = 80m}$$

In $\triangle BCD$, $\tan \beta = \frac{80}{120}$

$$\beta = \tan^{-1} \left(\frac{80}{120} \right) \checkmark$$

Q11. From the top of a 12 m tree, the angles of elevation and depression of the top and bottom of a building are 30° and 45° , respectively. Find (a) the height of the building and (b) the distance between the tree and building. [8 marks]



b)

$$d = 12 \tan 45^\circ = 12m$$

$$d = ? \quad \tan 45^\circ = \frac{12\text{m}}{d}$$

$\triangle AED$

$$\Rightarrow d = 12\text{m}$$

a) height of Building

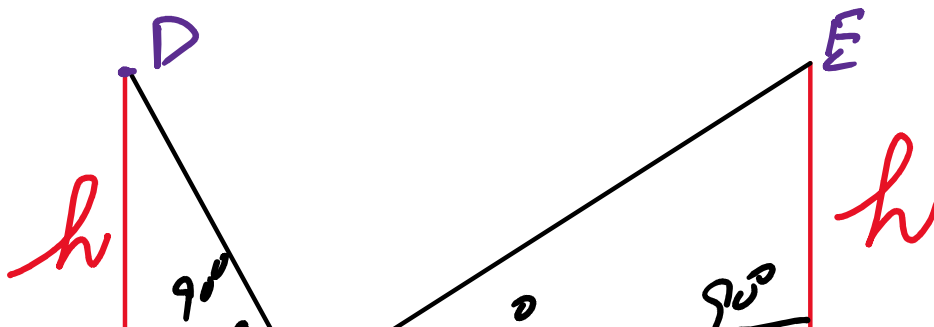
In $\triangle ABE$,

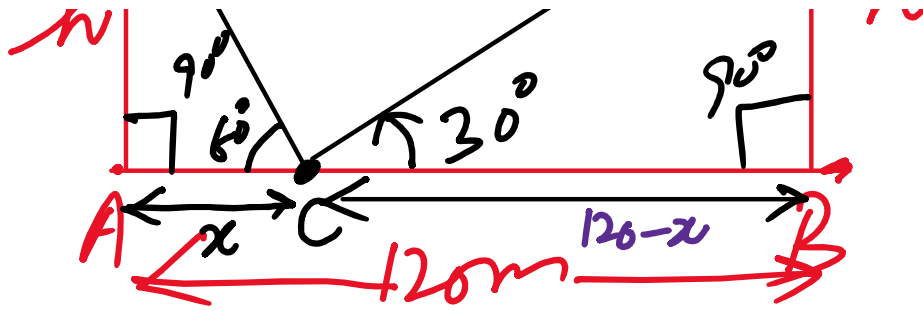
$$\tan 30^\circ = \frac{y}{12}$$

$$y = 12 \times \tan 30^\circ \text{ m}$$

$$\text{Height of Building} = y + 12\text{m}$$

Q12. Two pillars of equal height stand on opposite sides of a 120 m wide roadway. From a point between them, the angles of elevation of their tops are 60° and 30° . Find the height of each pillar and the position of the point. [8 marks]





In Rgt ΔACD

$$\tan 60^\circ = \frac{h}{x}$$

$$h = x\sqrt{3} \rightarrow (1)$$

In ΔBCE $\tan 30^\circ = \frac{h}{120-x}$

$$\Rightarrow h = \frac{120-x}{\sqrt{3}} \rightarrow (2)$$

As height of wire pillar

$$x\sqrt{3} = \frac{120-x}{\sqrt{3}}$$

$$3x = 120 - x$$

$$4x = 120$$

$$x = 30 \text{ m}$$

Thus Pillar A is 30m away
from the ~~marked~~ position + pin
is 90m away

$$\text{Eqn (1)} \Rightarrow h = 30\sqrt{3} \text{ m}$$

$$h \approx 52 \text{ m}$$

